

Pythagorean Proof Gallery Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 30-40 min

Category	Mathematics	Time	30-40 min
Difficulty	Medium	ID	pythagorean-proof-gallery

Big idea: Mathematical proof is not just a result; it is a standard for why a result must be true.

Student-Facing Prompt

Inspect Pythagorean Proof Gallery Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Pythagorean Proof Gallery Stimulus A	Student-facing stimulus 1: Visual rearrangement proof of $a^2 + b^2 = c^2$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Pythagorean Proof Gallery Stimulus B	Student-facing stimulus 2: Algebraic proof card plus a verbal explanation card of the same theorem. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Pythagorean Proof Gallery Reveal / Twist	Student-facing stimulus 3: Algebraic proof card plus a verbal explanation card of the same theorem. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Mathematical proof is not just a result; it is a standard for why a result must be true.

Actual Lesson Questions

#	Question for students
1	After inspecting Pythagorean Proof Gallery Stimulus A, what is your first knowledge claim?
2	Which exact detail from Pythagorean Proof Gallery Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	What makes mathematical proof different from empirical confirmation?

Ready-to-Use Examples

- Visual rearrangement proof of $a^2 + b^2 = c^2$.
- Algebraic proof card plus a verbal explanation card of the same theorem.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

1. Display three different proofs of $a^2 + b^2 = c^2$.
2. Groups annotate what each proof shows, assumes, and leaves unclear.
3. Students rank proofs by convincingness, explanatory power, and elegance.
4. Groups compare whether their rankings changed across criteria.
5. Debrief why mathematical knowledge values proof rather than repeated measurement alone.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: What makes a proof convincing?
- Can a visual proof explain better than an algebraic proof?
- Is elegance a mathematical virtue or an aesthetic judgement?

Knowledge Questions

- What makes mathematical proof different from empirical confirmation?
- How do different representations support different kinds of understanding?
- Does beauty play a role in mathematical knowledge?

Transfer Example

For example, run this activity with a simple classroom version using three proof cards: visual rearrangement, algebraic derivation, and verbal explanation.. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...